



MOHAMMED PREMIER UNIVERSITY
FACULTY OF SCIENCES – OUDJA
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Prepared by:

Pr. I. Jerrari

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1 Review on Groups

1.1 Basic Definitions

Definition 1. Let G be a set equipped with an internal binary operation $*$. We say that $(G, *)$ is a *group* if the following conditions hold:

1. (Associativity) For all $x, y, z \in G$,

$$(x * y) * z = x * (y * z).$$

2. (Identity element) There exists $e \in G$ such that for all $x \in G$,

$$e * x = x * e = x.$$

3. (Inverse element) For every $x \in G$, there exists $y \in G$ such that

$$x * y = y * x = e.$$

If moreover $x * y = y * x$ for all $x, y \in G$, then $(G, *)$ is called an *abelian group*.

Definition 2. A subset H of a group G is called a *subgroup* of G if:

- it contains the identity element,
- for all $x, y \in H$, both $x * y$ and x^{-1} belong to H .

Examples of Groups and Subgroups

1. $(\mathbb{Z}, +)$ is an abelian group and every subgroup is of the form $n\mathbb{Z}$.
2. If $x \in G$, the set $\{x^n \mid n \in \mathbb{Z}\}$ is a cyclic subgroup generated by x .
3. If G and H are groups, then the Cartesian product $G \times H$ equipped with

$$(g, h)(g', h') = (g * g', h \circ h')$$

is a group.

1.2 Order of Elements and Finite Groups

Definition 3. A group G is called *finite* if it has finitely many elements. The number of elements of G is called the *order* of G and is denoted $|G|$.

Definition 4. Let $x \in G$. If there exists a positive integer n such that $x^n = e$, the smallest such integer is called the *order of x* .

Examples.

1. $\mathbb{Z}/n\mathbb{Z}$ is a finite group and has order n .
2. The symmetric group S_n is non-abelian for $n > 2$ and has order $n!$.
3. If $ab = ba$, then

$$\text{ord}(ab) = \frac{\text{ord}(a) \text{ord}(b)}{\text{gcd}(\text{ord}(a), \text{ord}(b))}.$$

1.3 Lagrange's Theorem

Theorem 1 (Lagrange's Theorem). *Let G be a finite group and H a subgroup of G . Then $|H|$ divides $|G|$.*

Proof. Consider the relation xRy if and only if $y \in Hx$. This is an equivalence relation and each equivalence class has cardinality $|H|$. Since G is a disjoint union of equivalence classes, $|H|$ divides $|G|$. $\square$

1.4 Normal Subgroups

Let H be a subgroup of G .

Definition 5. The subgroup H is said to be *normal* in G if

$$\forall z \in G, zHz^{-1} = H.$$

Equivalently,

$$\forall z \in G, zH = Hz.$$

1.5 Group Homomorphisms

Definition 6. Let G and G' be two groups and $f : G \rightarrow G'$ a map. We say that f is a *group homomorphism* if

$$f(xy) = f(x)f(y)$$

for all $x, y \in G$. If f is bijective, it is called a *group isomorphism*. If $G = G'$, it is called an *automorphism*.

Definitions 1. Let $f : G \rightarrow G'$ be a homomorphism and let e' be the identity element of G' . The set

$$\ker(f) = \{x \in G \mid f(x) = e'\}$$

is called the *kernel* of f .

The set

$$\text{Im}(f) = \{f(x) \mid x \in G\}$$

is called the *image* of f .

Proposition 1. *Let $f : G \rightarrow G'$ be a homomorphism. Then:*

1. $\text{Im}(f)$ is a subgroup of G' .
2. $\ker(f)$ is a normal subgroup of G .

Proof. We prove that $\text{Im}(f)$ is a subgroup of G' . Let $y_1 = f(x_1)$ and $y_2 = f(x_2)$ be in $\text{Im}(f)$. Then

$$y_1 y_2^{-1} = f(x_1) f(x_2)^{-1} = f(x_1) f(x_2^{-1}) = f(x_1 x_2^{-1}),$$

which belongs to $\text{Im}(f)$. Hence $\text{Im}(f)$ is a subgroup of G' .

The fact that $\ker(f)$ is normal in G follows from

$$f(gxg^{-1}) = \text{Im}(f)f(x)\text{Im}(f)^{-1},$$

and if $x \in \ker(f)$, then $f(x) = e'$, hence

$$f(gxg^{-1}) = e'.$$

Thus $gxg^{-1} \in \ker(f)$. □

Examples.

1. The map

$$f : (\mathbb{C}, +) \rightarrow (\mathbb{C}^*, \cdot), \quad x \mapsto e^x$$

is a group homomorphism.

2. Let $f : S_n \rightarrow \{1, -1\}$ be defined by

$$f(\sigma) = (-1)^r$$

where σ is written as a product of r transpositions. Then $A_n = \ker(f)$ and A_n is normal in S_n .

3. If H is normal in G and $a \in G$, then

$$f_a(x) = axa^{-1}$$

defines an automorphism of H .

2 Quotient Groups

2.1 Quotient Sets

Let E and F be two sets, $f : E \rightarrow F$ a map, and R an equivalence relation on E .

Definition 7. The set of equivalence classes modulo R is called the *quotient set* of E by R and is denoted by E/R . The map

$$p : E \rightarrow E/R, \quad x \mapsto \bar{x}$$

is called the *canonical projection*.

Proposition 2. *With the previous notation, there exists a map*

$$\tilde{f} : E/R \rightarrow F$$

such that

$$f = \tilde{f} \circ p$$

if and only if

$$xRy \Rightarrow f(x) = f(y).$$

Proof. Assume that $f = \tilde{f} \circ p$. If xRy , then $\bar{x} = \bar{y}$, hence

$$f(x) = \tilde{f}(\bar{x}) = \tilde{f}(\bar{y}) = f(y).$$

Conversely, let $\bar{y} \in E/R$. Define

$$\tilde{f}(\bar{y}) = f(y).$$

The map $\tilde{f}$ is well defined since

$$\bar{x} = \bar{y} \iff xRy \implies f(x) = f(y).$$

□

Definition 8. The relation R is said to be *compatible* with f if

$$xRy \Rightarrow f(x) = f(y).$$

Definition 9. For every map $f : E \rightarrow F$, we define a relation R_f on E by

$$xR_f y \iff f(x) = f(y).$$

This is the equivalence relation associated with f .

Corollary 1. *There exists an injective map*

$$\tilde{f} : E/R_f \rightarrow F$$

such that $f = \tilde{f} \circ p$.

Proof. The map $\tilde{f}$ exists by Proposition 2. Moreover, we have

$$\bar{x} = \bar{y} \iff xR_f y \iff f(x) = f(y) \iff \tilde{f}(\bar{x}) = \tilde{f}(\bar{y}).$$

Hence $\tilde{f}$ is injective.

□

2.2 Quotient Groups

Let G be a group and H a subgroup of G . Define

$$xR_ry \iff y \in Hx \quad \text{and} \quad xR_\ell y \iff y \in xH.$$

Definition 10. If H is normal in G , then R_r and R_ℓ coincide. In this case, the quotient sets G/R_r and G/R_ℓ coincide and we define

$$G/H = \{xH \mid x \in G\}.$$

Proposition 3. If H is normal in G , then G/H equipped with

$$(xH)(yH) = xyH$$

is a group called the quotient group.

Proof. Assume that H is normal in G .

* **Well-defined law.** Suppose $xH = x'H$ and $yH = y'H$. Then there exist $h_1, h_2 \in H$ such that

$$x' = xh_1 \quad \text{and} \quad y' = yh_2.$$

Hence

$$x'y' = (xh_1)(yh_2) = x(h_1y)h_2.$$

Since H is normal, $h_1y = yh_3$ for some $h_3 \in H$. Thus

$$x'y' = xy(h_3h_2).$$

Because $h_3h_2 \in H$, we obtain

$$x'y'H = xyH.$$

Therefore the operation $(xH)(yH) = xyH$ is well defined.

* **Associativity.** For all $x, y, z \in G$,

$$\left((xH)(yH)\right)(zH) = (xyH)(zH) = (xy)zH = x(yz)H = (xH)(yzH) = (xH)\left((yH)(zH)\right),$$

since the operation in G is associative.

* **Identity element.** The class $H = eH$ satisfies

$$(xH)(H) = xH \quad \text{and} \quad (H)(xH) = xH$$

for all $x \in G$. Thus H is the identity element.

* **Inverse element.** For $x \in G$,

$$(xH)(x^{-1}H) = xx^{-1}H = eH = H.$$

Hence the inverse of xH is $x^{-1}H$.

Therefore G/H is a group. □

Definition 11. An equivalence relation R on G is compatible with the group law if

$$xRy \Rightarrow xzRyz \text{ and } zxRzy.$$

This is also equivalent to saying that for all $(x, y) \in G^2$ and for all $(x', y') \in G^2$,

$$xRy \text{ and } x'Ry' \Rightarrow xx'Ryy'.$$

Proposition 4. *An equivalence relation R on G is compatible with the group law if and only if the class of the identity element is a normal subgroup H of G and*

$$xRy \iff y \in xH.$$

Proof. Suppose that R is compatible with the group law on G , i.e.

$$xRy \text{ and } x'Ry' \Rightarrow xx'Ryy'.$$

Let H be the equivalence class of the identity element e :

$$H = \{x \in G \mid xRe\}.$$

* **H is a subgroup of G .** If $a, b \in H$, then aRe and bRe . Since R is compatible with the group law, we obtain

$$ab^{-1}Ree^{-1} = e,$$

hence $ab^{-1} \in H$. Therefore H is a subgroup of G .

* **H is normal in G .** Let $h \in H$ and $g \in G$. Since hRe and gRg , compatibility gives

$$ghg^{-1}Rgeg^{-1} = e,$$

so $ghg^{-1} \in H$. Hence H is a normal subgroup of G .

* **Description of the equivalence classes.** Let $x, y \in G$ such that xRy , then by compatibility

$$e = x^{-1}xRx^{-1}y,$$

hence $x^{-1}y \in H$, which implies $y \in xH$.

Conversely, if $y \in xH$, then $y = xh$ for some $h \in H$. Since hRe and xRx , compatibility yields

$$xhRxe = x,$$

thus yRx , and by symmetry xRy .

Therefore

$$xRy \iff y \in xH.$$

Conversely, assume that H is a normal subgroup of G and define the relation

$$xRy \iff y \in xH.$$

This relation is clearly an equivalence relation, and for xRy and $x'Ry'$ we have

$$y = xh, \quad y' = x'h'$$

with $h, h' \in H$. Then

$$yy' = x(hx')h'.$$

Since H is normal, $hx' = x'k$ for some $k \in H$, hence

$$yy' = xx'(kh') \in xx'H.$$

Thus $xx'Ryy'$, showing that R is compatible with the group law. □

2.3 First Isomorphism Theorem

Theorem 2 (First Isomorphism Theorem). *Let G and G' be two groups, f a homomorphism from G to G' , p the canonical surjection from G onto $G/\ker(f)$, and i the canonical injection from $\text{Im}(f)$ into G' . Then there exists an isomorphism $\bar{f}$ from $G/\ker(f)$ onto $\text{Im}(f)$ such that*

$$f = i \circ \bar{f} \circ p.$$

Proof. $G/\ker(f)$ is the quotient set of G by the relation R defined by

$$xRy \iff xy^{-1} \in \ker(f) \iff f(x) = f(y).$$

Thus R is the equivalence relation associated with f . Consequently, by Corollary 1, there exists an injective map f_1 such that $f = f_1 \circ p$, where

$$f_1 : G/\ker(f) \longrightarrow G', \quad \bar{x} \longmapsto f(x).$$

Let $\bar{f}$ be the map from $G/\ker(f)$ to $\text{Im}(f)$ defined by $\bar{f}(\bar{x}) = f(x)$. Then $\bar{f}$ is a group isomorphism and $f_1 = i \circ \bar{f}$. Hence

$$f = i \circ \bar{f} \circ p.$$

□

2.4 Factorization Property

Proposition 5. *Let $f : G \rightarrow G'$ and $g : G \rightarrow G''$ be homomorphisms with g surjective. Then there exists $\varphi : G'' \rightarrow G'$ such that*

$$f = \varphi \circ g$$

if and only if

$$\ker(g) \subset \ker(f).$$

Proof. The first implication is obvious. Conversely, let $y \in G''$. Since g is surjective, there exists $x \in G$ such that $y = g(x)$. Define

$$\varphi(y) = f(x).$$

Let us show that φ is well defined. If $y = g(x) = g(x')$, then

$$g(x)g(x')^{-1} = e'' = g(xx'^{-1}).$$

Hence $xx'^{-1} \in \ker(g)$. Since $\ker(g) \subseteq \ker(f)$, we obtain

$$f(xx'^{-1}) = e'.$$

Thus $f(x) = f(x')$, and therefore $\varphi(y) = \varphi(y')$. This proves that φ is well defined. Finally, we verify that φ is a group homomorphism. If $y = g(x)$ and $y' = g(x')$, then

$$\varphi(yy') = f(xx') = f(x)f(x') = \varphi(y)\varphi(y').$$

□

2.5 Products of Subgroups

Proposition 6. *Let H, K be subgroups of G .*

1. *HK is a subgroup if and only if $HK = KH$.*
2. *If H is normal, then HK is a subgroup.*
3. *If H is normal, then $H \cap K$ is normal in K .*

Proof.

1. Suppose that HK is a subgroup of G . Let us show that $HK = KH$.

Let $h \in H$ and $k \in K$. Since HK is a subgroup, we have

$$(hk)^{-1} = k^{-1}h^{-1} \in HK.$$

Hence there exist $h' \in H$ and $k' \in K$ such that

$$k^{-1}h^{-1} = h'k'.$$

Taking inverses gives

$$hk = k'^{-1}h'^{-1} \in KH.$$

Thus $HK \subseteq KH$.

Now let $kh \in KH$. Then

$$h^{-1}k^{-1} \in HK.$$

Taking inverses again gives

$$kh = (h^{-1}k^{-1})^{-1} \in HK.$$

Hence $KH \subseteq HK$, and therefore $HK = KH$.

Conversely, suppose $HK = KH$. Let $hk, h'k' \in HK$. Then

$$a = hk(h'k')^{-1} = hk(k'^{-1}h'^{-1}) = hkk'^{-1}h'^{-1}.$$

Let $k'' = kk'^{-1} \in K$. Since $HK = KH$, there exist $h'' \in H$ and $k''' \in K$ such that

$$k''h'^{-1} = h''k'''.$$

Thus

$$a = hh''k''' \in HK.$$

Hence HK is a subgroup of G .

2. If H is normal in G , then HK is a subgroup of G .

Indeed, it suffices to show that $HK = KH$. Since H is normal, for every $k \in K$ we have

$$kH = Hk.$$

Therefore $HK = KH$, and by the first part HK is a subgroup of G .

3. Let $x \in H \cap K$ and $k \in K$. Then

$$kxk^{-1} \in K,$$

and since H is normal in G , we also have

$$kxk^{-1} \in H.$$

Thus

$$kxk^{-1} \in H \cap K.$$

Therefore $H \cap K$ is a normal subgroup of K .

□

2.6 Second Isomorphism Theorem

Theorem 3 (Second Isomorphism Theorem). *Let H be a normal subgroup of G and K a subgroup of G . Then*

$$K/(H \cap K) \simeq HK/H.$$

Proof. Since H is a normal subgroup of G , it follows that HK is a subgroup of G . Hence H is a normal subgroup of HK . Consider the map

$$p : K \longrightarrow HK/H, \quad x \longmapsto xH.$$

The map p is a surjective homomorphism. Therefore,

$$K/\ker(p) \simeq HK/H.$$

Moreover,

$$\ker(p) = \{x \in K \mid xH = H\} = \{x \in K \mid x \in H\} = K \cap H.$$

Hence,

$$K/(K \cap H) \simeq HK/H.$$

□

2.7 Third Isomorphism Theorem

Theorem 4 (Third Isomorphism Theorem). *Let $H \subset K$ be normal subgroups of G . Then*

$$G/K \simeq (G/H)/(K/H).$$

Proof. Let p be the canonical surjection from G onto G/H and p' the canonical surjection from G onto G/K . Since $\ker(p) \subseteq \ker(p')$, it follows from Proposition 5 that there exists a map

$$\varphi : G/H \rightarrow G/K$$

such that

$$p' = \varphi \circ p.$$

The map φ is defined by sending xH to xK . It is clear that φ is surjective. Moreover,

$$\ker(\varphi) = \{xH \in G/H \mid xK = K\} = \{xH \mid x \in K\} = K/H.$$

Hence K/H is a normal subgroup of G/H and

$$(G/H)/(K/H) \simeq G/K.$$

□

2.8 Correspondence Theorem

Theorem 5 (Correspondence Theorem). *Let H be a normal subgroup of G . There is a bijection between subgroups of G/H and subgroups of G containing H .*

Proof. Let

$$\pi : G \longrightarrow G/H, \quad \pi(x) = xH$$

be the canonical surjection. Define

$$\Phi : \{K \leq G \mid H \subseteq K\} \longrightarrow \{\text{subgroups of } G/H\}$$

by

$$\Phi(K) = K/H.$$

First, if K is a subgroup of G containing H , then K/H is a subgroup of G/H , so Φ is well defined.

* **Injectivity.** Suppose K_1 and K_2 are subgroups of G containing H such that

$$\Phi(K_1) = \Phi(K_2).$$

Then

$$K_1/H = K_2/H.$$

Let $x \in K_1$. Then $xH \in K_1/H$. Since $K_1/H = K_2/H$, we have

$$xH \in K_2/H.$$

Hence there exists $y \in K_2$ such that

$$xH = yH.$$

Thus

$$y^{-1}x \in H.$$

Since $H \subseteq K_2$ and $y \in K_2$, and because K_2 is a subgroup, we obtain

$$x = y(y^{-1}x) \in K_2.$$

Therefore $K_1 \subseteq K_2$. Similarly $K_2 \subseteq K_1$, so $K_1 = K_2$.

* **Surjectivity.** Let T be a subgroup of G/H and define

$$K = \pi^{-1}(T) = \{x \in G \mid xH \in T\}.$$

Then K is a subgroup of G containing H , and

$$\Phi(K) = K/H = T.$$

Therefore Φ is a bijection between the subgroups of G/H and the subgroups of G containing H . $\square$

Application

The subgroups of $\mathbb{Z}/n\mathbb{Z}$ are in bijection with the subgroups H of $\mathbb{Z}$ such that $n\mathbb{Z} \subseteq H$. Since every subgroup of $\mathbb{Z}$ is of the form $d\mathbb{Z}$ for some $d \in \mathbb{N}$, we obtain that the subgroups of $\mathbb{Z}/n\mathbb{Z}$ correspond to the subgroups $d\mathbb{Z}$ of $\mathbb{Z}$ with $n\mathbb{Z} \subseteq d\mathbb{Z}$. Moreover, if $n = md$ with $\gcd(m, d) = 1$, we have

$$d\mathbb{Z}/n\mathbb{Z} \simeq \mathbb{Z}/m\mathbb{Z}.$$

3 Supplement on Groups

3.1 Group Acting on a Set

Definition 12. Let G be a group and E a set. An *action* of G on E is a map

$$\cdot : G \times E \rightarrow E, \quad (g, x) \mapsto g \cdot x$$

such that:

1. $\forall (g_1, g_2) \in G^2, \forall x \in E, (g_1 g_2) \cdot x = g_1 \cdot (g_2 \cdot x),$
2. $\forall x \in E, e \cdot x = x.$

We say that G acts on the left on E . In this case, E is called a G -set.

Proposition 7. *Let G be a group and E a set. Then E is a G -set if and only if there exists a group homomorphism $T : G \rightarrow S(E)$.*

Proof.

- If $G \subseteq S(E)$, define a natural action of G on E by:

$$G \times E \rightarrow E, \quad (g, x) \mapsto g \cdot x = g(x).$$

Then:

$$- \forall (g_1, g_2) \in G^2, \forall x \in E,$$

$$(g_1 \circ g_2) \cdot x = (g_1 \circ g_2)(x) = g_1(g_2(x)) = g_1 \cdot (g_2 \cdot x),$$

$$- \text{Id}_E \cdot x = \text{Id}_E(x) = x.$$

- Suppose there exists a homomorphism $T : G \rightarrow S(E)$. Define:

$$G \times E \rightarrow E, \quad (g, x) \mapsto g \cdot x = T(g)(x).$$

Then:

$$- \forall (g_1, g_2) \in G^2, \forall x \in E,$$

$$(g_1 g_2) \cdot x = T(g_1 g_2)(x) = (T(g_1) \circ T(g_2))(x) = T(g_1)(T(g_2)(x)) = g_1 \cdot (g_2 \cdot x),$$

$$- \forall x \in E, \quad e \cdot x = T(e)(x) = \text{Id}_E(x) = x.$$

Thus this defines an action, and E is a G -set.

- Conversely, suppose that G acts on E . Define:

$$T : G \rightarrow S(E), \quad g \mapsto T_g : E \rightarrow E, \quad T_g(x) = g \cdot x.$$

- T_g is bijective. Indeed, if $g \cdot x = g \cdot x'$, then

$$g^{-1} \cdot (g \cdot x) = g^{-1} \cdot (g \cdot x') \Rightarrow x = x'.$$

Thus T_g is injective. Moreover,

$$\forall y \in E, \quad y = g \cdot (g^{-1} \cdot y),$$

so T_g is surjective.

- T is a group homomorphism:

$$T(g_1 g_2)(x) = (g_1 g_2) \cdot x = g_1 \cdot (g_2 \cdot x) = (T_{g_1} \circ T_{g_2})(x).$$

Hence,

$$T(g_1 g_2) = T_{g_1} \circ T_{g_2}.$$

□

Definition 13. If T is injective, the action is called *faithful*.

Examples.

1. Let $E = \{1, 2, 3, \dots, n\}$ and $G = S_n$, the symmetric group of E . Then G acts on E , and the action is defined by:

$$G \times E \rightarrow E, \quad (\sigma, p) \mapsto \sigma(p).$$

2. Let G be a group and N a normal subgroup of G . Then G acts on N as follows:

$$G \times N \rightarrow N, \quad (g, n) \mapsto gng^{-1}.$$

3. Let G be a group. Then G acts on itself by conjugation:

$$G \times G \rightarrow G, \quad (g, x) \mapsto gxg^{-1}.$$

4. Let G be a group and A the set of all subgroups of G . Then G acts on A by:

$$G \times A \rightarrow A, \quad (g, N) \mapsto gNg^{-1}.$$

Definition 14 (Orbit of an element). Let E be a G -set. Define a relation T on E by:

$$xTy \iff y \in G \cdot x = \{g \cdot x \mid g \in G\}.$$

Then T is an equivalence relation. The equivalence class of x modulo T is called the *orbit* of x , denoted O_x , and we have:

$$O_x = G \cdot x$$

Example. Consider the case where G acts on itself by conjugation. Then for every x ,

$$O_x = \{gxg^{-1} \mid g \in G\}.$$

If the element x commutes with every element of G , then

$$O_x = \{x\}.$$

Definition 15 (Center of a group). Let G be a group. The set of elements of G that commute with every element of G is a subgroup of G , called the *center* of G , and is denoted by $Z(G)$.

Definition 16 (Stabilizer of an element). Let E be a G -set and $x \in E$. The set

$$P_x = \{g \in G \mid g \cdot x = x\}$$

is a subgroup of G , called the *stabilizer* of x .

Examples.

1. If G acts on itself by conjugation, then

$$P_x = \{g \in G \mid g \cdot x = x\} = \{g \in G \mid gxg^{-1} = x\} = \{g \in G \mid gx = xg\}.$$

In this case, P_x is also called the *centralizer* of x . Moreover, we have

$$Z(G) = \bigcap_{x \in G} P_x.$$

2. If G acts on A (the set of all subgroups of G) by conjugation, then

$$P_N = \{g \in G \mid gNg^{-1} = N\},$$

and P_N is called the *normalizer* of N .

3. Let S_4 be the group of permutations of $\{1, 2, 3, 4\}$, and let P_4 be the stabilizer of 4:

$$P_4 = \{\sigma \in S_4 \mid \sigma(4) = 4\}.$$

It is clear that $P_4 \simeq S_3$.

Theorem 6. *Let E be a G -set and $x \in E$. Then there exists a bijection between the orbit of x and the set of left cosets modulo P_x .*

Proof. Let $(G/P_x)_\ell$ be the set of left cosets of P_x , and define

$$\varphi : (G/P_x)_\ell \rightarrow O_x = G \cdot x$$

by

$$\varphi(gP_x) = g \cdot x$$

We have:

$$gP_x = g'P_x \iff g'^{-1}g \in P_x \iff g'^{-1}g \cdot x = x \iff g \cdot x = g' \cdot x$$

Thus φ is well-defined and injective. Moreover, it is surjective by construction: for every $y \in G \cdot x$, there exists $g \in G$ such that $y = g \cdot x$, hence

$$\varphi(gP_x) = y.$$

Therefore, φ is a bijection. □

Corollary 2. *Assume moreover that G is finite; then for every x , the orbit O_x is finite and $\text{card}(O_x)$ divides the order of G .*

Proof. If G is finite, then $(G/P_x)_\ell$ is finite; hence O_x is finite and $\text{card}(O_x) = [G : P_x]$. Therefore, $\text{card}(O_x)$ divides $|G|$. □

Corollary 3. *Let X be a set consisting of one representative from each orbit. Assume that E is a finite G -set. Then*

$$|E| = \sum_{x \in X} [G : P_x].$$

Proof. This follows from the fact that

$$E = \bigcup_{x \in X} O_x \Rightarrow |E| = \sum_{x \in X} |O_x| = \sum_{x \in X} [G : P_x].$$

□

3.2 Applications

3.2.1 Cauchy's Theorem

Theorem 7 (Cauchy's Theorem). *Let G be a group of order n and let p be a prime dividing n . Then there exists an element $x \in G$ whose order is equal to p .*

Proof. First, let

$$E = \{(x_1, x_2, \dots, x_p) \in G^p \mid x_1 x_2 \cdots x_p = e\}.$$

We will show that $|E| = n^{p-1}$. To this end, define

$$f : E \rightarrow G^{p-1}, \quad (x_1, x_2, \dots, x_p) \mapsto (x_1, x_2, \dots, x_{p-1}).$$

Clearly, f is well-defined. Suppose that

$$f(x_1, \dots, x_p) = f(x'_1, \dots, x'_p),$$

then

$$(x_1, \dots, x_{p-1}) = (x'_1, \dots, x'_{p-1}),$$

so

$$x_1 = x'_1, x_2 = x'_2, \dots, x_{p-1} = x'_{p-1}.$$

Since $x_1 \cdots x_p = e$ and $x'_1 \cdots x'_p = e$, it follows that $x_p = x'_p$. Hence f is injective. Moreover, f is surjective since for every $(x_1, \dots, x_{p-1}) \in G^{p-1}$,

$$f(x_1, \dots, x_{p-1}, x_p) = (x_1, \dots, x_{p-1}),$$

where

$$x_p = (x_1 x_2 \cdots x_{p-1})^{-1}.$$

Thus $|E| = |G^{p-1}| = n^{p-1}$.

Now, define

$$\sigma : E \rightarrow E, \quad (x_1, x_2, \dots, x_p) \mapsto (x_2, x_3, \dots, x_p, x_1).$$

Then σ is injective, hence bijective since E is finite. Moreover,

$$\sigma^p = \text{id}_E.$$

Thus the order of σ is either 1 or p .

– If $\text{ord}(\sigma) = 1$, then $\sigma = \text{id}_E$, so for all $y \in E$, $\sigma(y) = y$.

Let

$$y = (x_1, x_2, \dots, x_p) \neq (e, e, \dots, e).$$

Then

$$(x_2, x_3, \dots, x_p, x_1) = (x_1, x_2, \dots, x_p),$$

hence

$$x_1 = x_2 = \cdots = x_p = x \neq e.$$

Since $(x_1, \dots, x_p) \in E$, we have $x^p = e$, so x has order p .

- If $\text{ord}(\sigma) = p$, let

$$C = \{\text{id}_E, \sigma, \sigma^2, \dots, \sigma^{p-1}\} \subset S(E).$$

Then C acts naturally on E , hence

$$|E| = \sum_{x \in X} |O_x|,$$

where each orbit size $|O_x|$ divides $|C| = p$.

Thus $|O_x| = 1$ or p . Suppose there are r orbits of size 1 and s orbits of size p . Then

$$|E| = r + sp = n^{p-1}.$$

Since

$$O_{(e, \dots, e)} = C \cdot (e, \dots, e) = \{(e, \dots, e)\},$$

we have $r \geq 1$.

Thus $r + sp = n^{p-1}$ implies $p \mid r$, hence $r \geq p$. Therefore, there exists

$$(x_1, \dots, x_p) \in E, \quad (x_1, \dots, x_p) \neq (e, \dots, e),$$

such that its orbit has size 1. Hence

$$(x_2, x_3, \dots, x_p, x_1) = (x_1, x_2, \dots, x_p),$$

so

$$x_1 = x_2 = \dots = x_p = x \neq e.$$

Since $(x_1, \dots, x_p) \in E$, we get $x^p = e$, hence there exists an element of order p .

□

Definition 17 (p -group). Let G be a group and p a prime number. We say that G is a p -group if and only if every element of G has order a power of p .

Proposition 8. *Let G be a finite group. Then G is a p -group if and only if the order of G is a power of p .*

Proof. Let $n = |G|$ and let q be a prime divisor of n . By Cauchy's Theorem, there exists $x \in G$ such that the order of x is q . But by hypothesis, every element of G has order a power of p . Hence $\text{ord}(x) = p^i$ for some integer i , so $q = p^i$. Since q is prime, it follows that $q = p$. Therefore, p is the only prime divisor of $|G|$, and thus $|G|$ is a power of p . The converse is obvious. □

3.2.2 Class Equation

Let G be a finite group. Consider the action of G on itself by conjugation. Then we have:

$$|G| = \sum_{x \in X} [G : P_x].$$

Moreover,

$$[G : P_x] = 1 \iff G = P_x \iff gxg^{-1} = x, \forall g \in G \iff x \in Z(G).$$

If $x \in Z(G)$, then its orbit is

$$O_x = \{gxg^{-1} \mid g \in G\} = \{x\},$$

hence $Z(G) \subseteq X$, where X is a set of representatives of the orbits.

Thus,

$$|G| = \sum_{x \in Z(G)} [G : P_x] + \sum_{x \notin Z(G)} [G : P_x] = |Z(G)| + \sum_{x \notin Z(G)} [G : P_x].$$

This last equation is called the *class equation*.

Theorem 8 (Burnside's Theorem). *Let G be a finite p -group. Then the center of G is nontrivial.*

Proof. We know that $[G : P_x]$ divides $|G| = p^n$ from Lagrange's Theorem. Moreover, if $x \notin Z(G)$, then $[G : P_x] \neq 1$, hence $p \mid [G : P_x]$. It follows that

$$p \mid \sum_{x \notin Z(G)} [G : P_x].$$

Since $p \mid |G|$, we conclude from the class equation that $p \mid |Z(G)|$. Therefore, $Z(G)$ contains at least p elements, and in particular is not reduced to the identity element. $\square$

3.2.3 Sylow's Theorems

Let G be a group of order n , and let d be a divisor of n . Does there exist a subgroup of G of order d ? In what follows, we will try to answer this question.

Theorem 9 (Sylow's First Theorem). *Let G be a group of order $n = p^h m$, where p is a prime number such that $p \nmid m$. Then for every $k \leq h$, there exists a subgroup of G of order p^k .*

Proof. We argue by induction on n . For $n = 1, 2, 3, 4$, the result is true. Assume that the result holds for all integers $i < n$, and let us prove it for n .

We have the class equation:

$$|G| = |Z(G)| + \sum_{x \notin Z(G)} [G : P_x].$$

- Suppose that $p \nmid |Z(G)|$. Then there exists $y \notin Z(G)$ such that $p \nmid [G : P_y]$. Hence $p^k \nmid [G : P_y]$. By Lagrange's Theorem, we have

$$|G| = |P_y| \cdot [G : P_y]$$

Since $p^k \mid |G|$ and $p^k \nmid [G : P_y]$, it follows that $p^k \mid |P_y|$.

Now $|P_y| \neq n$, otherwise $[G : P_y] = 1$, which implies $y \in Z(G)$, a contradiction. Hence $|P_y| < n$. By the induction hypothesis, there exists a subgroup of P_y , hence of G , of order p^k .

- Suppose that $p \mid |Z(G)|$. By Cauchy's Theorem, there exists an element $a \in Z(G)$ of order p . Let $H = \langle a \rangle$. Then H is a normal subgroup of G , since $a \in Z(G)$.

Thus G/H is a group whose order is strictly smaller than that of G . Since $p^k \mid |G|$ and $|H| = p$, it follows that $p^{k-1} \mid |G/H|$. By the induction hypothesis, there exists a subgroup T of G/H of order p^{k-1} .

By the Correspondence Theorem, there exists a subgroup R of G containing H such that $T = R/H$. Hence

$$|R| = |T| \cdot |H| = p^k,$$

and therefore there exists a subgroup of G of order p^k .

□

Definition 18. A p -Sylow subgroup of G is a subgroup of G of order p^h , where p^h is the largest power of p dividing $|G|$.

Remark 1. A p -Sylow subgroup is a maximal element in the set of p -subgroups of G . Indeed, if H is a p -Sylow subgroup and K is a p -subgroup of G such that $H \subseteq K$, then we have $|K| = p^i$ and $|H| = p^h$. Since $H \subseteq K$, it follows that $h \leq i$. But h is the largest integer such that $p^h \mid |G|$, hence $i = h$. Therefore, $H = K$.

Theorem 10 (Sylow's Second Theorem). Let G be a group of order n , and let p be a prime divisor of n .

1. If H is a p -Sylow subgroup of G , then for every $a \in G$, the subgroup aHa^{-1} is also a p -Sylow subgroup.
2. If H is the unique p -Sylow subgroup of G , then H is normal in G .

Proof.

1. Let

$$\varphi_a : H \rightarrow aHa^{-1}, \quad x \mapsto axa^{-1}.$$

Then φ_a is a group automorphism. Hence, the subgroups H and aHa^{-1} have the same order. Therefore,

$$|H| = |aHa^{-1}| = p^h,$$

the largest power of p dividing n . It follows that for every $a \in G$, the subgroup aHa^{-1} is a p -Sylow subgroup of G .

2. If H is the unique p -Sylow subgroup, then by 1), for every $a \in G$, we have

$$H = aHa^{-1},$$

and thus H is normal in G . □

Lemma 1. *Let H be a p -Sylow subgroup of G . Then H is the unique element of $N(H)/H$ whose order is a power of p .*

Proof. Since

$$N(H) = \{g \in G \mid gHg^{-1} = H\},$$

it is clear that H is normal in $N(H)$. We will show that if aH has order a power of p , then $aH = H$.

Let T be the p -subgroup of $N(H)/H$ generated by aH . By the Correspondence Theorem, there exists a subgroup R of $N(H)$ containing H such that $T = R/H$. Since T is a p -group, it follows that R is a p -subgroup of G . Therefore, since $H \subseteq R$ and H is maximal among the p -subgroups of G , we obtain $H = R$. Hence,

$$T = H/H = \{H\},$$

and thus $aH = H$. □

Lemma 2. *Let H be a p -Sylow subgroup of G , and let $a \in G$ be an element of order a power of p such that $aHa^{-1} = H$. Then $a \in H$.*

Proof. Since $aHa^{-1} = H$, we have $a \in N(H)$. As a has order a power of p , it follows that aH has order a power of p in $N(H)/H$. But by the previous lemma, H is the only element of $N(H)/H$ whose order is a power of p . Hence $aH = H$, and therefore $a \in H$. □

Theorem 11 (Sylow's Third Theorem). *Let G be a group of order $n = p^h m$, where p is a prime number such that $p \nmid m$. Then:*

1. *The number of p -Sylow subgroups of G is congruent to 1 modulo p and divides $|G|$.*
2. *All p -Sylow subgroups of G are conjugate.*

Proof.

1. Let H be a p -Sylow subgroup of G and let

$$A = \{H, H_1, H_2, \dots, H_r\}$$

be the set of all conjugates of H . Then G acts on A by conjugation:

$$G \times A \rightarrow A, \quad (g, H_i) \mapsto gH_i g^{-1}.$$

Similarly, if K is a p -subgroup of G , then K acts on A in the same way, and the cardinality of each orbit divides $|K|$. Taking $K = H$, we have

$$|A| = \sum_{x \in X} |O_x|,$$

and each $|O_x|$ divides $|H| = p^h$.

- * $O_H = \{gHg^{-1} \mid g \in H\} = \{H\}$, so O_H contains exactly one element.
- * Let $H_i \in A$ with $H_i \neq H$. We claim that $O_{H_i} \neq \{H_i\}$. Otherwise, if

$$O_{H_i} = \{H_i\},$$

then for all $a \in H$, we have $aH_i a^{-1} = H_i$. Since H_i is a p -Sylow subgroup and $a \in H$ has order a power of p , it follows from Lemma 2 that $a \in H_i$. Hence $H \subseteq H_i$. Since both are p -Sylow subgroups, they have the same order, so $H = H_i$, a contradiction.

Therefore,

$$|A| = 1 + p^{\alpha_1} + p^{\alpha_2} + \cdots + p^{\alpha_s},$$

with $\alpha_i \neq 0$, hence

$$|A| \equiv 1 \pmod{p}.$$

Now let H_0 be a p -Sylow subgroup and suppose that $H_0 \notin A$. Then H_0 acts on A , and the orbit sizes divide $|H_0|$. We show that no orbit has size 1. Otherwise, suppose there exists $K \in A$ such that

$$O_K = \{K\}.$$

Then for all $a \in H_0$, we have $aKa^{-1} = K$, hence $H_0 \subseteq K$ (from Lemma 2), which implies $H_0 = K$, a contradiction.

Thus all orbits have size greater than 1, so

$$|A| = p^{\beta_1} + p^{\beta_2} + \cdots + p^{\beta_t},$$

with $\beta_i \neq 0$, hence

$$|A| \equiv 0 \pmod{p},$$

a contradiction. Therefore, $H_0 \in A$.

It follows that A is the set of all p -Sylow subgroups of G , and thus their number is congruent to 1 modulo p .

Finally, define

$$f : A \rightarrow (G/N(H))_\ell, \quad aHa^{-1} \mapsto aN(H).$$

Let a and b be elements of G . Then we have the following equivalences:

$$aHa^{-1} = bHb^{-1} \iff b^{-1}aHa^{-1}b = H \iff b^{-1}a \in N(H) \iff aN(H) = bN(H).$$

Hence, $aHa^{-1} = bHb^{-1} \iff f(aHa^{-1}) = f(bHb^{-1})$, and therefore f is injective. Moreover, f is surjective by construction. Thus f is bijective, and therefore

$$|A| = [G : N(H)].$$

2. Let H_i and H_j be two p -Sylow subgroups of G . Then $H_i, H_j \in A$, the set of all conjugates of a fixed p -Sylow subgroup H . Hence, there exist $a, b \in G$ such that

$$H_i = aHa^{-1}, \quad H_j = bHb^{-1}.$$

Equivalently,

$$b^{-1}H_jb = H.$$

It follows that

$$H_i = aHa^{-1} = a(b^{-1}H_jb)a^{-1} = (ab^{-1})H_j(ab^{-1})^{-1}.$$

Therefore, H_i and H_j are conjugate, and hence all p -Sylow subgroups of G are conjugate. □

Remark 2. The number of p -Sylow subgroups, denoted by N_p , divides m . Indeed; by Sylow's Third Theorem, we know that:

- $N_p \equiv 1 \pmod{p}$,
- $N_p \mid |G|$.

Since $N_p \equiv 1 \pmod{p}$, there exists an integer k such that

$$N_p = 1 + kp.$$

If $p \mid N_p$, then $N_p \equiv 0 \pmod{p}$, which contradicts $N_p \equiv 1 \pmod{p}$. Therefore, $p \nmid N_p$. Since $N_p \mid |G| = p^h m$ and $\gcd(N_p, p) = 1$, it follows that $N_p \mid m$.

Theorem 12 (Sylow's Fourth Theorem). *Let G be a finite group and let p be a prime divisor of $|G|$. Then every p -subgroup of G is contained in a p -Sylow subgroup of G .*

Proof. Let K be a p -subgroup of G , let H be a p -Sylow subgroup of G , and let

$$A = \{H = H_0, H_1, H_2, \dots, H_r\}$$

be the set of all p -Sylow subgroups of G . Then K acts on A by conjugation. Since

$$|A| = \sum_{x \in X} |O_x|,$$

where O_x denotes the orbit of x , and since $|O_x|$ divides $|K| = p^s$, it follows that $|O_x|$ is a power of p . Moreover, $|A| \equiv 1 \pmod{p}$, hence there exists an index i such that $|O_{H_i}| = 1$. This means that

$$O_{H_i} = \{gH_i g^{-1} \mid g \in K\} = \{H_i\}.$$

Thus, for all $a \in K$, we have

$$aH_i a^{-1} = H_i.$$

By Lemma 2, this implies that

$$K \subseteq H_i.$$

□

Theorem 13 (Sylow's Fifth Theorem). *Let G be a finite group and let p be a prime divisor of $|G|$. Then the following are equivalent:*

- H is a p -Sylow subgroup of G ,
- H is maximal in the set of p -subgroups of G .

Proof. The first implication was proved in Remark 1. By Sylow's Fourth Theorem, every p -subgroup K of G is contained in a p -Sylow subgroup H of G . If moreover K is maximal, then it must coincide with the p -Sylow subgroup H . Hence the converse implication follows. $\square$

Example. Let G be a group of order $6 = 2 \times 3$, and let N_p denote the number of p -Sylow subgroups of G .

- (a) If G is abelian, then $N_2 = N_3 = 1$, since every subgroup of G is normal in G .
- (b) If $N_2 = N_3 = 1$, then G has two proper subgroups: a 2-Sylow subgroup and a 3-Sylow subgroup. These two subgroups are normal, and their product is G . This implies that G is abelian.
- (c) If G is non-abelian, then since $N_p \equiv 1 \pmod{p}$, we have $N_2 = 1$ or 3 , and $N_3 = 1$. If $N_2 = 1$, then since $N_3 = 1$, the group G would be abelian, which contradicts our assumption. Hence, $N_2 = 3$. This implies in particular that there exists a unique 3-Sylow subgroup, and it is normal in G .
If $G = S_3$, the symmetric group, then $N_2 = 3$, since there are three 2-Sylow subgroups of S_3 , namely those generated by the three transpositions.

3.3 Solvable Groups

3.3.1 Derived Series of a Group

We denote by $D(G)$ the group $[G, G]$ generated by the commutators of G , also called the *derived subgroup* of G . We define recursively on n the n th derived subgroup $D^n(G)$ by

$$D^n(G) = D(D^{n-1}(G)).$$

Properties

1. Let G and G' be two groups and let

$$\varphi : G \longrightarrow G'$$

be a group homomorphism. Then

$$\varphi(D(G)) \subseteq D(G').$$

If φ is surjective, then

$$\varphi(D(G)) = D(G').$$

By induction, one can verify that

$$\varphi(D^n(G)) \subseteq D^n(G').$$

Moreover, if φ is surjective, then

$$\varphi(D^n(G)) = D^n(G').$$

2. Let H be a subgroup of G and let

$$i : H \longrightarrow G$$

be the canonical injection. By Property (1), we have

$$i(D^n(H)) = D^n(H) \subseteq D^n(G).$$

3. Assume furthermore that H is a normal subgroup of G . Consider

$$p : G \longrightarrow G/H, \quad x \longmapsto \bar{x}.$$

Since p is surjective, we have

$$p(D^n(G)) = D^n(G/H).$$

Moreover,

$$p(D^n(G)) = HD^n(G)/H.$$

Remark 3. Since

$$\varphi(D(G)) \subseteq D(G')$$

for every homomorphism φ , in particular when $G' = G$ and φ is an inner automorphism, the inclusion still holds. Hence $D(G)$ is a normal subgroup of G . Similarly, each derived subgroup

$$D^n(G) = D(D^{n-1}(G))$$

is normal in $D^{n-1}(G)$.

3.3.2 Normal Series of a Group

Let G be a group and

$$G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_k \supseteq \cdots \supseteq G_n = \{e\}$$

a sequence of subgroups of G . The sequence (G_i) is said to be a *normal series* if and only if G_{i+1} is a normal subgroup of G_i for every $0 \leq i \leq n-1$.

Example. Let S_4 be the symmetric group on $\{1, 2, 3, 4\}$, let A_4 be the subgroup of even permutations, and let

$$V_4 = \{\sigma \in A_4 \mid \sigma^2 = \text{id}\}.$$

Then

$$V_4 \triangleleft A_4 \quad \text{and} \quad A_4 \triangleleft S_4.$$

Hence

$$S_4 \supseteq A_4 \supseteq V_4 \supseteq \{e\}$$

is a normal series.

3.3.3 Solvable Groups

A group G is said to be *solvable* if and only if there exists a normal series

$$G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_k \supseteq \cdots \supseteq G_n = \{e\}$$

such that each quotient G_i/G_{i+1} is abelian for all $0 \leq i \leq n-1$.

Examples.

1. The group S_4 is solvable since

$$S_4/A_4, \quad A_4/V_4, \quad \text{and} \quad V_4/\{e\}$$

are all abelian.

2. Every abelian group is solvable.

Theorem 14. *Let G be a group. Then G is solvable if and only if there exists an integer n such that*

$$D^n(G) = \{e\}.$$

Proof. Suppose that G is solvable. Then there exists a normal series

$$G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_n = \{e\}$$

such that each quotient G_i/G_{i+1} is abelian. In particular, G/G_1 is abelian, hence

$$D(G) \subseteq G_1.$$

Since G_1/G_2 is abelian, we obtain

$$D(G_1) \subseteq G_2.$$

Therefore,

$$D^2(G) = D(D(G)) \subseteq D(G_1) \subseteq G_2.$$

By induction, one proves that

$$D^i(G) \subseteq G_i \quad \text{for all } i.$$

Thus

$$D^n(G) \subseteq G_n = \{e\},$$

which implies

$$D^n(G) = \{e\}.$$

Conversely, assume that there exists n such that

$$D^n(G) = \{e\}.$$

Then

$$G = D^0(G) \supseteq D(G) \supseteq D^2(G) \supseteq \cdots \supseteq D^n(G) = \{e\}$$

is a normal series. Moreover,

$$D^i(G)/D^{i+1}(G) = D^i(G)/D(D^i(G))$$

is abelian. Hence G is solvable. □

Theorem 15. *Let G be a group and let H be a normal subgroup of G . Then G is solvable if and only if both H and G/H are solvable.*

Proof. Assume that G is solvable. Then there exists an integer n such that

$$D^n(G) = \{e\}.$$

Since

$$D^n(H) \subseteq D^n(G) = \{e\},$$

the subgroup H is solvable. Moreover,

$$D^n(G/H) = HD^n(G)/H = \{H\},$$

hence G/H is solvable.

Conversely, assume that H and G/H are solvable. Then there exists an integer n such that

$$D^n(G/H) = \{H\},$$

and an integer m such that

$$D^m(H) = \{e\}.$$

Since

$$D^n(G/H) = HD^n(G)/H = \{H\},$$

it follows that

$$D^n(G) \subseteq H.$$

Therefore,

$$D^{m+n}(G) = D^m(D^n(G)) \subseteq D^m(H) = \{e\}.$$

Thus G is solvable. □

Remark 4. Let G be a finite p -group, then G is solvable. Indeed, we argue by induction on n , where $|G| = p^n$.

- For $n = 0, 1, 2$, the result is obvious, since in these cases G is abelian and hence solvable.
- Assume that the result holds for every integer $i < n$. We prove that G is solvable.
 - If $G = Z(G)$, then G is abelian and therefore solvable.
 - If $G \neq Z(G)$, then $Z(G)$ is solvable (because it is abelian) and

$$G/Z(G)$$

has order p^k with $k < n$ (because $Z(G) \neq \{e\}$ by Burnside's Theorem). By the induction hypothesis,

$$G/Z(G)$$

is solvable. Consequently, G is solvable.

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